Modeling and Control of a PV-Powered Industrial Cold Storage System

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1 Introduction and System Modeling

1.1 Problem Definition

The objective of this project is the modeling and control of an industrial cold-storage unit for the temporary storage of temperature-sensitive products. Examples include processed food, pharmaceutical semi-finished products, chemical intermediates, or precision materials that must remain below a given temperature threshold in order to preserve product quality.

The storage system is equipped with the following components:

- A refrigeration unit with variable cooling power.
- An internal air-circulation fan.
- A photovoltaic (PV) generation system.
- A battery energy-storage system.
- An electrical connection to the external grid.

The main control objectives are to regulate the product temperature, which depends on the cold-room temperature, and to manage the available electrical energy so as to maximize the use of PV power and reduce grid consumption.

1.2 State Variables and Outputs

The system is characterized by three state variables:

$$x(t) = \begin{bmatrix} T_a(t) \\ T_p(t) \\ S_b(t) \end{bmatrix}$$

where:

- $T_a(t)$: Cold-room air temperature [°C].

- $T_p(t)$: Product temperature [°C].
- $S_b(t)$: Battery state of charge [p.u.].

Since all state variables are directly measurable, the measured output vector is:

$$y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix} = \begin{bmatrix} T_a(t) \\ T_p(t) \\ S_b(t) \end{bmatrix}$$

corresponding respectively to a cold-room air-temperature sensor, a product-temperature sensor, and a battery state-of-charge measurement. Thus $g(x(t)) = x(t)$.

1.3 Manipulated Inputs

To achieve the control objectives, there are three manipulated inputs:

$$u(t) = \begin{bmatrix} u_r(t) \\ u_f(t) \\ u_b(t) \end{bmatrix}$$

- $u_r(t)$: Refrigeration-power command [p.u.], with $0 \leq u_r(t) \leq 1$.
- $u_f(t)$: Internal fan-speed command [p.u.], with $0 \leq u_f(t) \leq 1$.
- $u_b(t)$: Battery power command [kW], with $-4 \leq u_b(t) \leq 4$. (Note: $u_b(t) > 0$ corresponds to battery discharge to the load, while $u_b(t) < 0$ corresponds to battery charging.)

1.4 External Disturbances

The cold-storage system is affected by three external disturbances:

$$d(t) = \begin{bmatrix} T_e(t) \\ P_{PV}(t) \\ Q_p(t) \end{bmatrix}$$

- $T_e(t)$: Outside ambient temperature [°C], modeled as:

$$T_e(t) = 24 + 8\sin\left(\frac{2\pi}{24}(t-8)\right)$$

- $P_{PV}(t)$: PV electrical power available to the system [kW], given by:

$$P_{PV}(t) = P_{\max}\frac{R_s(t)}{R_{\max}}$$

where the solar-radiation profile is:

$$R_s(t) = \begin{cases} R_{\max} \sin\left(\pi \dfrac{t - t_{sr}}{t_{ss} - t_{sr}}\right), & t_{sr} \le t \le t_{ss} \\ 0, & \text{otherwise} \end{cases}$$

with $R_{\max} = 850$ W/m^2, $t_{sr} = 6$ h, $t_{ss} = 18$ h, and $P_{\max} = 9$ kW.

- $Q_p(t)$: Internal product-load disturbance [W], modeled as:

$$Q_p(t) = \begin{cases} 1200, & 0 \le t < 4 \\ 600, & 4 \le t < 10 \\ 250, & 10 \le t \le 24 \end{cases}$$

1.5 System Parameters

The numerical parameters of the model are summarized in Table 1.

Symbol	Description	Value	Unit
C_a	Thermal capacity of chamber air	1.5×10^5	J/°C
C_p	Thermal capacity of stored product	8.0×10^5	J/°C
U_w	Wall heat-transmission coefficient	3.8	W/(m^2 °C)
A_w	Equivalent wall exchange area	70	m^2
k_r	Nominal refrigeration gain	1.6×10^4	W
k_f	Fan-enhanced heat-transfer coefficient	450	W/°C
k_e	Ambient thermal disturbance coefficient	$U_w A_w = 266$	W/°C
E_b	Battery energy capacity	18	kWh
η_b	Battery charge/discharge efficiency	0.94	–
P_f	Nominal fan electrical power	1.5	kW
P_r	Nominal refrigeration electrical power	7.0	kW

Table 1: System parameters.

1.6 Nonlinear Continuous-Time Model

The continuous-time state-space model of the PV-powered cold-storage system is derived from physical considerations. The nonlinear dynamic equations describing the evolution of the state variables are formulated as follows:

Chamber Air Temperature Balance:

$$\dot{T}_a(t) = \frac{1}{C_a}\Big[k_e\big(T_e(t) - T_a(t)\big) - k_r u_r(t) - k_f\big(1 + \alpha_f u_f(t)\big)\big(T_a(t) - T_p(t)\big) + Q_p(t)\Big] \tag{1}$$

Product Temperature Balance:

$$\dot{T}_p(t) = \frac{1}{C_p}\Big[k_f\big(1 + \alpha_f u_f(t)\big)\big(T_a(t) - T_p(t)\big)\Big] \tag{2}$$

Battery State of Charge Balance:

$$\dot{S}_b(t) = \frac{1}{E_b}\left[\eta_b \max(-u_b(t), 0) - \frac{1}{\eta_b}\max(u_b(t), 0)\right] \tag{3}$$

The total electrical load required by the actuators is:

$$P_{\text{load}}(t) = P_r u_r(t) + P_f u_f(t) \tag{4}$$

Then, the power imported from the external grid is:

$$P_{\text{grid}}(t) = P_{\text{load}}(t) - P_{PV}(t) - u_b(t) \tag{5}$$

A positive value of $P_{\text{grid}}(t)$ indicates grid import, while a negative value indicates surplus energy.

1.7 Operating Constraints

The system must remain within the following admissible operating region at all times:

$$0 \leq T_a(t) \leq 12\,^\circ\text{C}, \qquad 0 \leq T_p(t) \leq 8\,^\circ\text{C}, \qquad 0.15 \leq S_b(t) \leq 0.95.$$

1.8 Dynamic Couplings Analysis

The following dynamic interactions are explicitly included in the model and must be addressed in the control design:

1. **Thermal Coupling (Air and Product):** Chamber air temperature and product temperature are dynamically coupled through the air-product thermal exchange term. Increasing the refrigeration power reduces the chamber air temperature, which indirectly cools the product.

2. **Actuator Trade-offs (Fan Speed):** Increasing the fan speed improves the heat exchange between air and product, thereby accelerating product cooling, but also increases electrical consumption.

3. **Disturbance Interventions:** The outside temperature acts as a thermal disturbance through the chamber walls trying to alter the product temperature.

4. **Electrical and Energy Couplings:** The battery state of charge depends on the electrical energy balance of the system and therefore depends on refrigeration power, fan usage, and PV generation.Obviously, Higher PV availability improves the sustainability of the operation by reducing grid import. Furthermore, the state equations and auxiliary power-balance equations are coupled, meaning that at least one control action affects more than one relevant variable in the process.

2 Continuous-Time PID Control Design

2.1 Control Architecture

The control strategy is based on three PID controllers operating in continuous time, each acting on one manipulated input. The chosen input-output pairings are:

- **PID 1** acts on $u_r(t)$ to track the chamber air temperature reference $T_{a,ref}(t)$;
- **PID 2** acts on $u_f(t)$ to improve the thermal exchange between air and product, helping $T_p(t)$ converge toward the desired final value;
- **Supervisory action** acts on $u_b(t)$ to keep $S_b(t)$ close to the nominal value $S_{b,nom} = 0.60$ while maximizing the use of available PV energy.

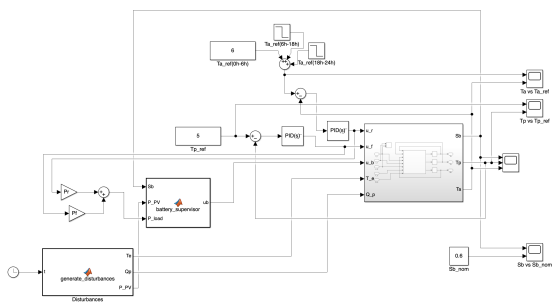

Figure 1: Continuous-time PID architecture.

2.2 PID Control Laws

The general continuous-time PID law for a control loop with tracking error $e(t) = r(t) - y(t)$ is:

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d\, \frac{de(t)}{dt} \tag{6}$$

where K_p, K_i, and K_d are respectively the proportional, integral, and derivative gains.

PID 1 — Chamber Air Temperature (u_r):

$$e_a(t) = T_{a,ref}(t) - T_a(t) \tag{7}$$

$$u_r(t) = \mathrm{sat}_{[0,1]}\left(K_{p,r}\, e_a(t) + K_{i,r} \int_0^t e_a(\tau)\, d\tau + K_{d,r}\, \frac{de_a(t)}{dt} \right) \tag{8}$$

PID 2 — Product Temperature (u_f):

$$e_p(t) = T_{p,ref} - T_p(t) \tag{9}$$

$$u_f(t) = \mathrm{sat}_{[0,1]}\left(K_{p,f}\, e_p(t) + K_{i,f} \int_0^t e_p(\tau)\, d\tau + K_{d,f}\, \frac{de_p(t)}{dt}\right) \tag{10}$$

Supervisory Battery Action (u_b):

The battery command is determined by a supervisory logic that combines a feedforward term based on the net power balance and a proportional feedback on the state of charge:

$$P_{net}(t) = P_{load}(t) - P_{PV}(t) \tag{11}$$

$$u_b(t) = \mathrm{sat}_{[-4,4]}(P_{net}(t) - K_b\,(S_{b,nom} - S_b(t))) \tag{12}$$

where $K_b > 0$ is the supervisory gain. A negative u_b (charging) is allowed only when a PV surplus is available, i.e. when $P_{net}(t) < 0$, in order to avoid drawing grid power to charge the battery.

2.3 Role of Each PID Term

For PID 1, the *proportional* term provides an immediate reaction to temperature deviations from the reference. The *integral* term eliminates the steady-state offset caused by persistent thermal disturbances (e.g., outside temperature T_e). For PID 2, the same structure applies, but with slower dynamics due to the larger thermal capacity $C_p \gg C_a$. The fan acts on the heat exchange coefficient rather than directly on $T_p(t)$, making this loop inherently slower and more coupled.

2.4 Tuned Gains

The gains were tuned heuristically based on the physical time constants of the system. The values used in the continuous-time simulation are reported in Table 2.

Controller	K_p	K_i	K_d
PID 1 (u_r)	-0.5	-1	0
PID 2 (u_f)	-0.5	-1	0
Supervisory (u_b)	$K_b = 100$	/	/

Table 2: Continuous-time PID gains.

2.5 Simulation Results

The continuous-time PID control was simulated over the 24-hour horizon with the initial conditions $T_a(0) = 10\,°\mathrm{C}$, $T_p(0) = 11.5\,°\mathrm{C}$, $S_b(0) = 0.55$. The results show that in a continuous model a perfect following of the constraints given by the assignment is achieved.

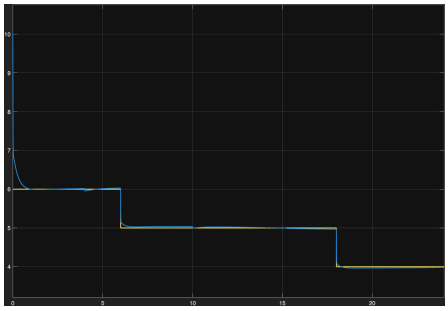

Figure 2: Chamber air temp $T_a(t)$ vs ref.

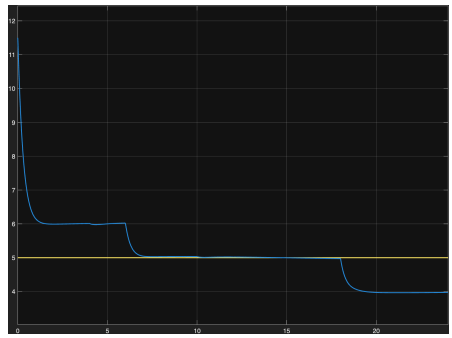

Figure 3: Product temp $T_p(t)$ vs target.

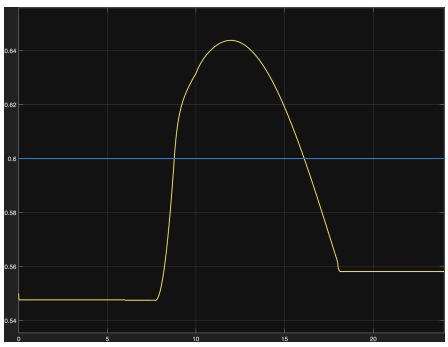

Figure 4: Battery SoC $S_b(t)$ vs nom.

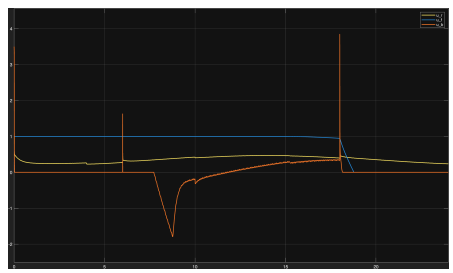

Figure 5: Control signals u_r, u_f, u_b.

3 Digital PID Control: ZOH and Euler Discretization

3.1 Digital PID with ZOH Applied to the Continuous-Time Plant

In this task, the PID controllers are implemented digitally with sampling time $T_s = 0.1$ h, while the plant remains continuous-time. The discrete control output is applied to the plant through a Zero-Order Hold (ZOH) block, which holds the control signal constant between two consecutive sampling instants.

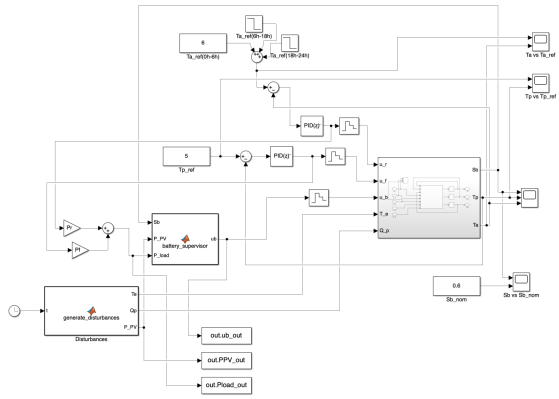

Figure 6: ZOH system architecture.

3.1.1 Discrete-Time PID Law

Loop 1 — Chamber air temperature (u_r):

$$e_{a,k} = T_{a,ref,k} - T_{a,k}, \qquad u_{r,k} = \mathrm{sat}_{[0,1]}\left(K_{p,r}\, e_{a,k} + K_{i,r}\, T_s \sum_{j=0}^{k} e_{a,j} + K_{d,r}\, \frac{e_{a,k} - e_{a,k-1}}{T_s} \right) \tag{13}$$

Loop 2 — Product temperature (u_f):

$$e_{p,k} = T_{p,ref} - T_{p,k}, \qquad u_{f,k} = \mathrm{sat}_{[0,1]}\left(K_{p,f}\, e_{p,k} + K_{i,f}\, T_s \sum_{j=0}^{k} e_{p,j} + K_{d,f}\, \frac{e_{p,k} - e_{p,k-1}}{T_s} \right) \tag{14}$$

Loop 3 — Battery supervisory action (u_b):

$$u_{b,k} = \mathrm{sat}_{[-4,4]}\left(P_{net,k} - K_b \left(S_{b,nom} - S_{b,k} \right) \right), \qquad P_{net,k} = P_{load,k} - P_{PV,k} \tag{15}$$

3.1.2 Simulation Results — ZOH Model

The hybrid system (discrete PID + ZOH + continuous nonlinear plant) was simulated over 24 hours with $T_s = 0.1$ h. The results are shown in Figures 7–10.

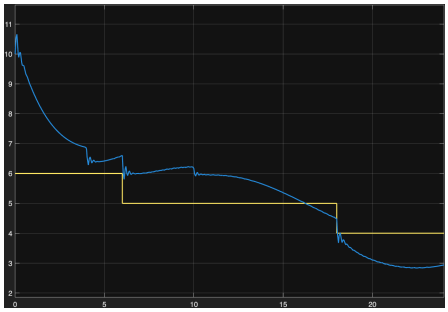

Figure 7: Chamber air temp $T_a(t)$ vs ref (ZOH).

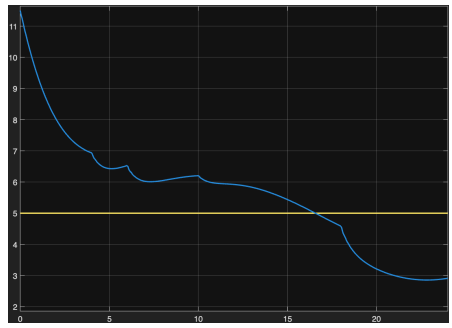

Figure 8: Product temp $T_p(t)$ vs target (ZOH).

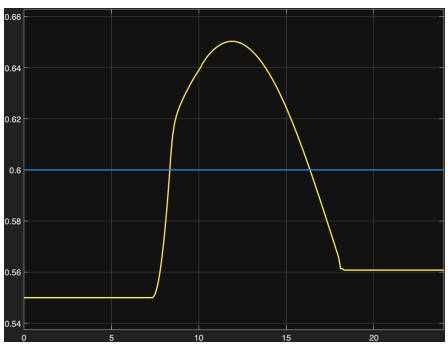

Figure 9: Battery SoC $S_b(t)$ vs nom (ZOH).

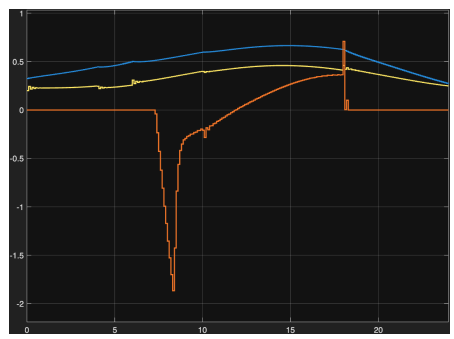

Figure 10: Control signals u_r, u_f, u_b (ZOH).

The ZOH model achieves excellent tracking of $T_{a,ref}(t)$ with minor transient oscillations at the reference step changes. The product temperature reaches $T_p(24) \approx 2.91\,^\circ\mathrm{C}$, well below the $5\,^\circ\mathrm{C}$ constraint. The battery state of charge follows the PV generation profile, charging during daylight hours and returning close to $S_{b,nom}$ by the end of the day, while remaining within the admissible bounds [0.15, 0.95] at all times.

3.2 Euler Discretization of the Nonlinear Model

In this task, the nonlinear continuous-time model is fully discretized using the Forward Euler method with sampling time T_s. The resulting discrete-time model takes the form $x_{k+1} = F(x_k, u_k, d_k)$:

$$T_{a,k+1} = T_{a,k} + T_s \cdot \frac{1}{C_a}\Big[k_e\big(T_{e,k} - T_{a,k}\big) - k_r u_{r,k} - k_f\big(1 + \alpha_f u_{f,k}\big)\big(T_{a,k} - T_{p,k}\big) + Q_{p,k}\Big] \tag{16}$$

$$T_{p,k+1} = T_{p,k} + T_s \cdot \frac{1}{C_p}\Big[k_f\big(1 + \alpha_f u_{f,k}\big)\big(T_{a,k} - T_{p,k}\big)\Big] \tag{17}$$

$$S_{b,k+1} = S_{b,k} + T_s \cdot \frac{1}{E_b}\Big[\eta_b \max(-u_{b,k}, 0) - \frac{1}{\eta_b}\max(u_{b,k}, 0)\Big] \tag{18}$$

PID design made with Tuning Simulink method. The obtained results are in the following table:

Controller	K_p	K_i	K_d	N
PID 1 (u_r)	0.01708	0.004314	0.01511	6.00
PID 2 (u_r)	0.01708	0.004314	0.01511	6.00
Supervisory (u_b)	$K_b = 100$	/	/	/

Table 3: Discrete-time PID gains for Euler model.

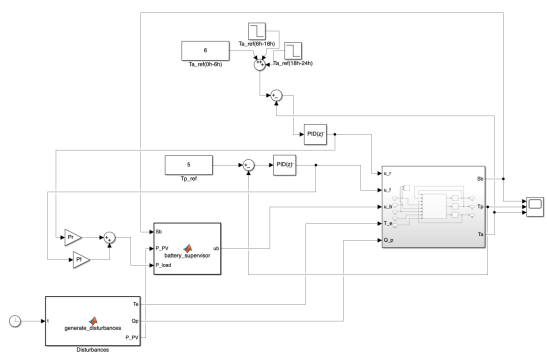

Figure 11: Euler system architecture.

3.2.1 Simulation Results — Euler Model

The Euler-discretized model was simulated with $T_s = 0.1$ h. As shown in Figures 12–15, the system exhibits numerical instability and diverges completely.

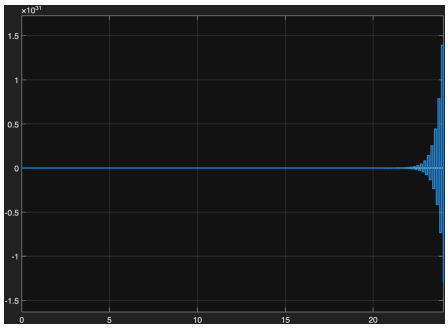

Figure 12: Chamber air temp $T_a(t)$ (Euler).

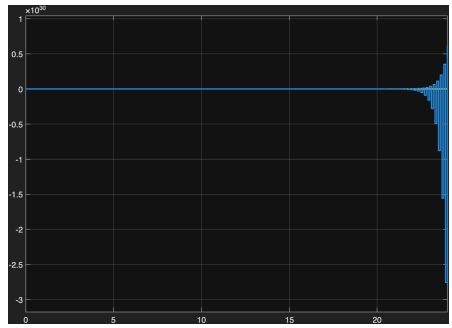

Figure 13: Product temp $T_p(t)$ (Euler).

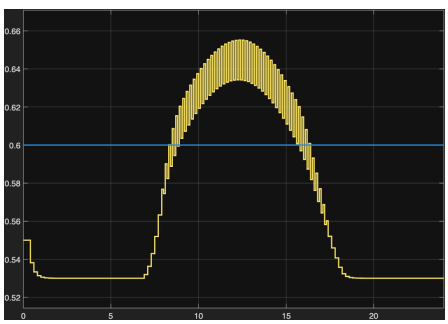

Figure 14: Battery SoC $S_b(t)$ (Euler).

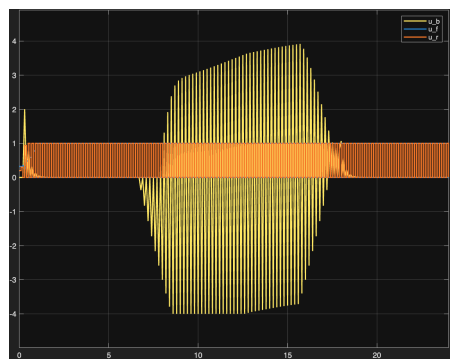

Figure 15: Control signals u_r, u_f, u_b (Euler).

3.3 Comparison: ZOH vs Euler at $T_s = 0.1$ h

Table 4 summarizes the key performance metrics for the two approaches at $T_s = 0.1$ h.

Table 4: Performance comparison between ZOH and Euler models at $T_s = 0.1$ h.

Metric	**ZOH + Nonlinear**	**Euler Nonlinear**
RMSE T_a [°C]	1.22	$\sim 10^{30}$ (diverges)
$T_p(24)$ [°C]	2.91	diverges
E_{grid} [kWh]	36.04	57.89
$E_{PV,use}$ [kWh]	39.71	39.46
PV fraction [%]	52.0	38.9
S_b within bounds	YES	YES (before divergence)
Numerical stability	**stable**	**unstable**

The ZOH approach demonstrates clear superiority: it maintains closed-loop stability, accurate temperature tracking, and efficient renewable energy utilization. The Euler model, despite sharing the same PID gains, fails due to the inherent stability limitation of explicit integration methods at this sampling time. This result confirms that the choice of discretization method has a critical impact not only on numerical accuracy but also on closed-loop stability and energy sustainability.

4 Equilibrium Point, Linearization, and Discretized Linear Model

4.1 Equilibrium Point and Linearization

4.1.1 Choice of Equilibrium Point

An equilibrium point $(x^\star, u^\star, d^\star)$ is determined by imposing $\dot{x}(t) = 0$ and using representative average values of the recipe references and external disturbances:

$$x^\star = \begin{bmatrix} T_a^\star \\ T_p^\star \\ S_b^\star \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 0.60 \end{bmatrix}, \qquad d^\star = \begin{bmatrix} T_e^\star \\ Q_p^\star \end{bmatrix} = \begin{bmatrix} 24 \\ 500 \end{bmatrix}, \qquad u^\star = \begin{bmatrix} u_r^\star \\ u_f^\star \\ u_b^\star \end{bmatrix} = \begin{bmatrix} 0.3471 \\ 0.5 \\ 0.0 \end{bmatrix}$$

The value $u_r^\star$ is computed analytically by imposing $\dot{T}_a = 0$:

$$u_r^\star = \frac{k_e(T_e^\star - T_a^\star) + Q_p^\star}{k_r} = \frac{266 \cdot (24 - 5) + 500}{1.6 \times 10^4} \approx 0.3471 \tag{19}$$

The fan is fixed at $u_f^\star = 0.5$ to maintain active air-product heat exchange at equilibrium, and the battery is at rest ($u_b^\star = 0$).

4.1.2 Linearized Continuous-Time Model

Defining the deviation variables $\delta x = x - x^\star$, $\delta u = u - u^\star$, $\delta d = d - d^\star$, the linearized model is:

$$\delta\dot{x}(t) = A\,\delta x(t) + B\,\delta u(t) + E\,\delta d(t), \qquad \delta y(t) = C\,\delta x(t) + D\,\delta u(t) \tag{20}$$

The Jacobian matrices are computed symbolically and evaluated at the equilibrium point. Since $T_a^\star = T_p^\star = 5\,°\mathrm{C}$, the coupling terms involving $(T_a - T_p)$ vanish in B:

$$A = \left.\frac{\partial f}{\partial x}\right|_\star = \begin{bmatrix} -\dfrac{k_e + k_f(1+\alpha_f u_f^\star)}{C_a/3600} & \dfrac{k_f(1+\alpha_f u_f^\star)}{C_a/3600} & 0 \\ \dfrac{k_f(1+\alpha_f u_f^\star)}{C_p/3600} & -\dfrac{k_f(1+\alpha_f u_f^\star)}{C_p/3600} & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -22.584 & 16.200 & 0 \\ 3.038 & -3.038 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{21}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\star} = \begin{bmatrix} -k_r \cdot \dfrac{3600}{C_a} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\dfrac{1}{E_b} \end{bmatrix} = \begin{bmatrix} -384.0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -0.0556 \end{bmatrix} \tag{22}$$

$$E = \left.\frac{\partial f}{\partial d}\right|_{\star} = \begin{bmatrix} k_e \cdot \dfrac{3600}{C_a} & \dfrac{3600}{C_a} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 6.384 & 0.024 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{23}$$

$$C = I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad D = \mathbf{0}_{3\times 3} \tag{24}$$

Physical interpretation of matrix A: The diagonal terms $A_{11} = -22.584$ and $A_{22} = -3.038$ represent the self-damping of the air and product temperatures respectively. The off-diagonal terms $A_{12} = 16.2$ and $A_{21} = 3.038$ represent the bidirectional thermal coupling between air and product through the fan-enhanced heat exchange. The zero third row confirms that S_b has no self-dynamics in the linearized model (the battery integrates power directly).

4.2 Discretization of the Linearized Model

The linearized continuous-time model is discretized using the exact Zero-Order Hold (ZOH) method with sampling time $T_s = 0.1$ h, via the MATLAB `c2d` function. The numerical values computed by MATLAB are:

$$A_d = \begin{bmatrix} 0.1623 & 0.5666 & 0 \\ 0.1062 & 0.8460 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix}, \quad B_d = \begin{bmatrix} -16.3050 & 0 & 0 \\ -2.8747 & 0 & 0 \\ 0 & 0 & -0.0056 \end{bmatrix} \tag{25}$$

$$E_d = \begin{bmatrix} 0.2711 & 0.0010 \\ 0.0478 & 0.0002 \\ 0 & 0 \end{bmatrix}, \qquad C_d = I_3, \qquad D_d = \mathbf{0}_{3\times 3} \tag{26}$$

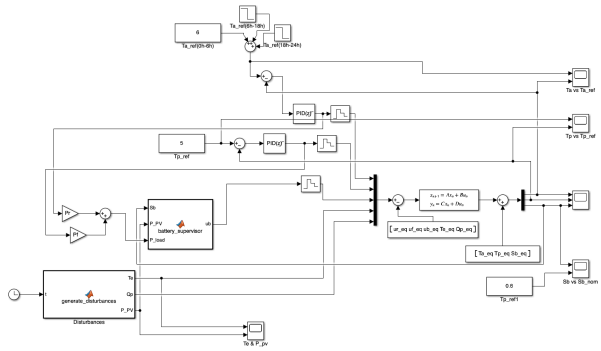

Figure 16: Linearized discrete model Architecture

4.3 Simulation Results — Linearized Discrete Model

The linearized discrete model achieves satisfactory temperature regulation, with $T_p(24) \approx 2.90\,^\circ\mathrm{C}$, well below the $5\,^\circ\mathrm{C}$ quality constraint, and $\mathrm{RMSE}_{T_a} \approx 1.22\,^\circ\mathrm{C}$. The battery state of charge correctly follows the PV generation profile, charging during daylight hours (6–18 h) and returning to approximately $S_b \approx 0.56$ at the end of the day, remaining within the admissible bounds $[0.15, 0.95]$ at all times.

The tracking performance is visually comparable to the ZOH model (Figure 7). However, as discussed in Section 5, the linearized model incurs significantly higher grid energy consumption ($E_{grid} \approx 49.7$ kWh vs 36.0 kWh for ZOH), because the controller must compensate for the mismatch between the linear approximation and the actual nonlinear plant. This effect is most pronounced during the initial cool-down transient, where the system operates far from the equilibrium point $x^\star = [5,\, 5,\, 0.60]^\top$.

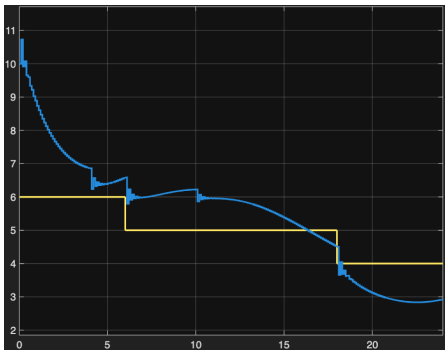

Figure 17: Chamber air temp $T_a(t)$ vs ref.

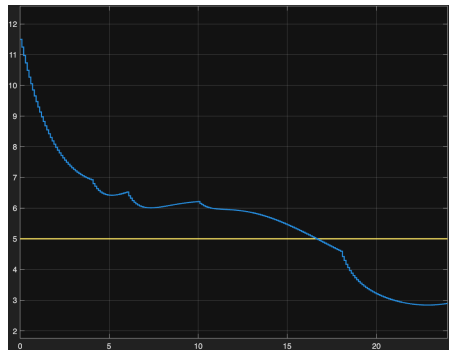

Figure 18: Product temp $T_p(t)$ vs target.

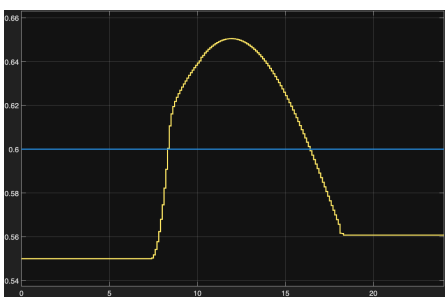

Figure 19: Battery SoC $S_b(t)$ vs $S_{b,nom}$.

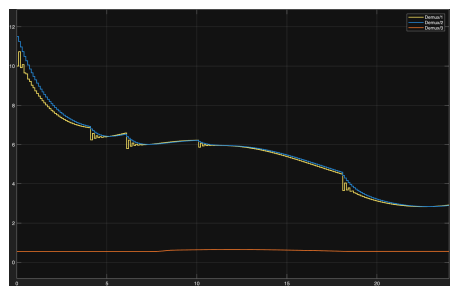

Figure 20: State variables T_a, T_p, S_b.

5 Effect of Sampling Time

This section analyzes the impact of the sampling time T_s on the control system's performance, numerical stability, and energy sustainability indicators. The analysis compares the three modeling approaches: ZOH + Nonlin, Euler Nonlin, and Lin + Disc, for $T_s \in \{0.05, 0.10, 0.25, 0.50\}$ h.

The overall simulation results are summarized in Table 5.

Table 5: Comparison of performance and sustainability indicators for different values of T_s.

Model	T_s **[h]**	**RMSE** T_a **[°C]**	$T_p(24)$ **[°C]**	E_{grid} **[kWh]**	$E_{PV,use}$ **[kWh]**	**PV [%]**	S_b **OK**
ZOH + Nonlin	0.05	1.2086	2.90	36.008	39.718	52.0%	TRUE
ZOH + Nonlin	0.10	1.2195	2.91	36.038	39.713	52.0%	TRUE
ZOH + Nonlin	0.25	1.2405	2.94	36.083	39.718	51.9%	TRUE
ZOH + Nonlin	0.50	1.2367	3.42	36.962	39.779	51.8%	TRUE
Euler Nonlin	0.05	21.1778	20.26	0.001	0.000	0.0%	TRUE
Euler Nonlin	0.10	12.1339	-11.93	58.853	41.334	39.5%	TRUE
Euler Nonlin	0.25	$\sim 1.40 \times 10^{66}$	$\sim 2.09 \times 10^{66}$	57.843	39.440	39.1%	TRUE
Euler Nonlin	0.50	$\sim 5.24 \times 10^{49}$	$\sim 5.50 \times 10^{49}$	56.864	39.368	39.3%	TRUE
Lin + Disc	0.05	1.1911	2.94	45.940	47.403	50.8%	TRUE
Lin + Disc	0.10	1.2224	2.90	49.706	48.809	49.5%	TRUE
Lin + Disc	0.25	1.2965	2.79	60.645	52.699	46.3%	TRUE
Lin + Disc	0.50	1.3825	2.81	77.621	58.438	42.5%	TRUE

5.1 External Disturbances

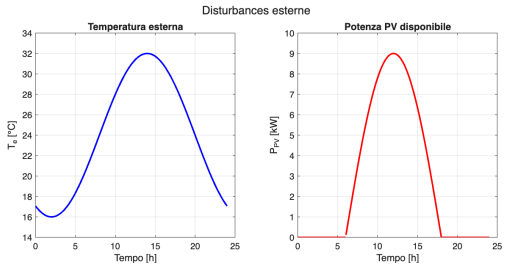

Figure 21: External disturbances: outside temperature $T_e(t)$ and available PV power $P_{PV}(t)$.

5.2 Per-Model Analysis Across Sampling Times

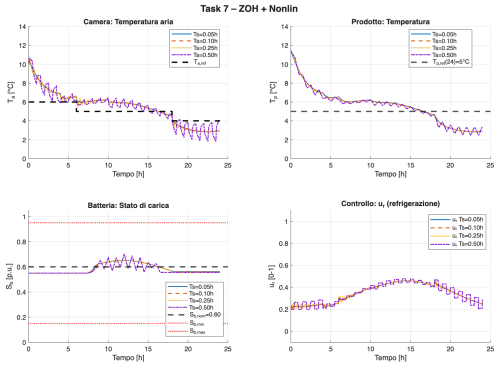

Figure 22: ZOH + Nonlin model: effect of sampling time on T_a, T_p, S_b, and u_r.

The ZOH + Nonlin model demonstrates remarkable robustness across all tested sampling times. The four curves for T_a are nearly indistinguishable, confirming that the continuous-time plant absorbs the discretization error introduced by the ZOH block. Only at $T_s = 0.50$ h do minor oscillations appear in the last hours, without compromising product quality. The refrigeration command u_r remains smooth and consistent across all T_s values.

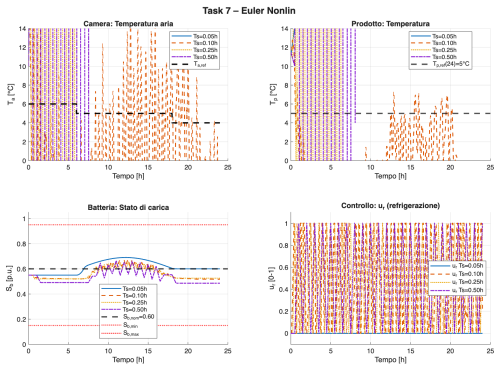

Figure 23: Euler Nonlin model: effect of sampling time on T_a, T_p, S_b, and u_r.

The Euler Nonlin model clearly shows the conditional stability limit of the Forward Euler method. Following the discrete-time PID retuning, the system no longer mathematically diverges at $T_s = 0.10$ h. However, it still fails to provide acceptable control: at $T_s = 0.05$ h the control effort is virtually zero (leading to unmitigated temperature rises), and at $T_s = 0.10$ h it exhibits violent, physically impossible oscillations (e.g., T_p dropping to $-11.9\,°\mathrm{C}$). True numerical divergence (values exceeding 10^{66}) is now delayed to $T_s \geq 0.25$ h. This confirms that while tuning can alter the exact divergence threshold, it cannot overcome the inherent stability limitations of the Euler method.

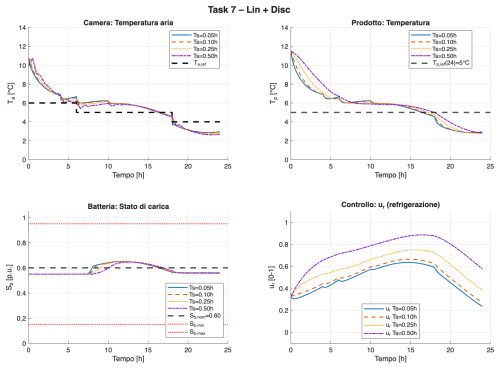

Figure 24: Lin + Disc model: effect of sampling time on T_a, T_p, S_b, and u_r.

The Lin + Disc model remains stable for all tested T_s values, but at a significant cost in control effort. As T_s increases, the refrigeration command u_r becomes progressively more aggressive, reaching saturation for longer periods. This directly translates into the higher grid energy consumption observed in Table 5.

5.3 Per-Sampling-Time Comparison Across Models

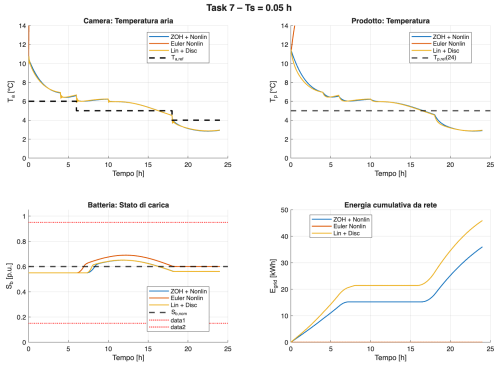

Figure 25: Comparison of the three models at $T_s = 0.05$ h.

At $T_s = 0.05$ h all three models avoid mathematical divergence. However, the retuned Euler model fails to control the system effectively: its control effort u_r remains negligible, causing a massive temperature deviation ($\mathrm{RMSE}_{T_a} \approx 21.2\,^{\circ}\mathrm{C}$) and zero PV utilization. ZOH and Lin + Disc, instead, track the reference smoothly. The cumulative grid energy E_{grid} strictly separates the latter two approaches: ZOH $\approx$ 36 kWh vs Lin + Disc $\approx$ 46 kWh.

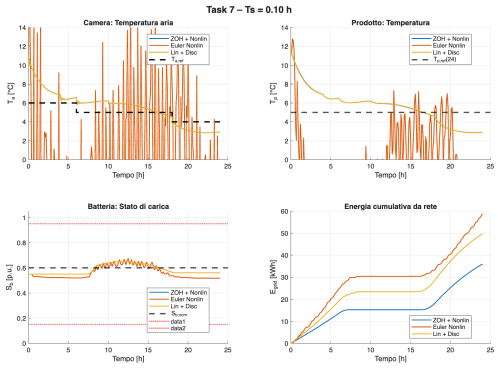

Figure 26: Comparison of the three models at $T_s = 0.10$ h.

At $T_s = 0.10$ h, the retuned Euler model avoids infinite values, but it breaks down into highly aggressive, physically unrealistic oscillations (with T_p reaching below $-11\,°\mathrm{C}$). ZOH and Lin + Disc both track correctly, but Lin + Disc consumes approximately 38% more grid energy than ZOH (≈ 50 kWh vs ≈ 36 kWh).

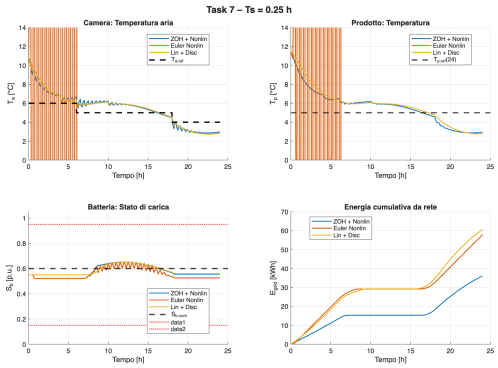

Figure 27: Comparison of the three models at $T_s = 0.25$ h.

At $T_s = 0.25$ h, the Euler model reaches its mathematical breaking point, diverging to values in the order of 10^{66}. The gap between ZOH and Lin + Disc in grid consumption widens further (≈ 36 kWh vs ≈ 61 kWh). The Lin + Disc model begins to show visible oscillations in T_a tracking at the reference step changes, a sign of increasing model-plant mismatch at lower sampling frequencies.

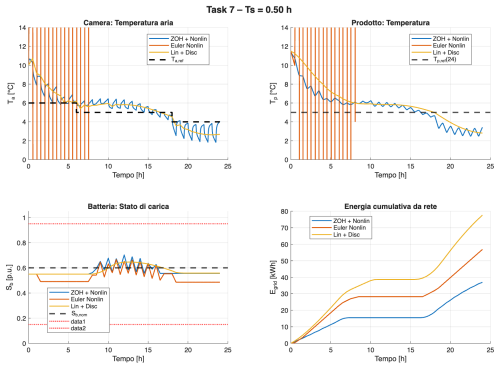

Figure 28: Comparison of the three models at $T_s = 0.50$ h.

At $T_s = 0.50$ h the ZOH model shows noticeable oscillations in the final hours but still satisfies all constraints. The Lin + Disc model reaches $E_{grid} \approx 78$ kWh, more than twice the ZOH value, with oscillations visible in both T_a and S_b.

5.4 Battery Command Comparison

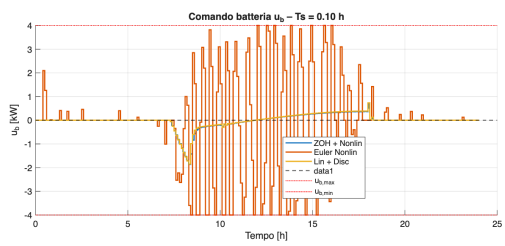

Figure 29: Battery command $u_b(t)$ for the three models at $T_s = 0.10$ h.

The ZOH and Lin + Disc models produce smooth, physically meaningful u_b profiles that follow the PV availability: charging (negative u_b) during daylight hours and discharging (positive u_b) when PV is unavailable. The Euler model generates highly oscillatory u_b commands that saturate continuously between -4 and $+4$ kW, making the battery supervisory logic completely ineffective.

5.5 Tracking Accuracy and Product Quality

The ZOH + Nonlin model demonstrates exceptional robustness. The RMSE_{T_a} varies negligibly from 1.21 to 1.24 °C across all T_s, and $T_p(24)$ remains safely below 5 °C in all cases. Only at $T_s = 0.50$ h is a slight degradation observed ($T_p = 3.42$ °C), without compromising product quality. The Lin + Disc model also maintains acceptable tracking, but at the cost of much higher control effort and grid consumption.

5.6 Numerical Stability and the Euler Limit

The Forward Euler method is only conditionally stable. The stability condition requires:

$$T_s < \frac{2}{|\lambda_{\max}|} \tag{27}$$

where $\lambda_{\max}$ is the dominant eigenvalue of the system Jacobian. The fast air thermal dynamics governed by C_a impose a tight stability bound. While the explicit PID retuning manages to prevent the immediate state explosion at $T_s = 0.10$ h, it cannot stabilize the actual control loop. Numerical divergence is simply delayed to $T_s \geq 0.25$ h, proving that the Euler method's structural limitations cannot be bypassed via controller tuning alone.

5.7 Sustainability and Grid Dependence

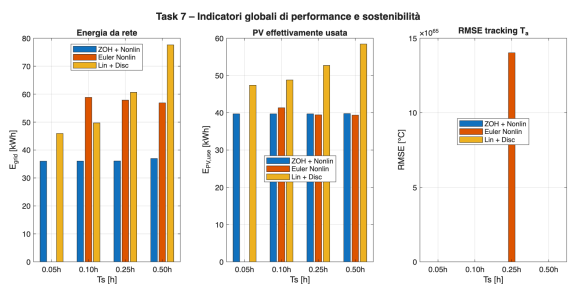

Figure 30: Global performance and sustainability indicators: grid energy E_{grid}, PV energy used $E_{PV,use}$, and RMSE_{T_a} for all models and sampling times.

The bar charts confirm three key findings. First, the ZOH model maintains a flat $E_{grid} \approx 36$ kWh regardless of T_s, with PV utilization consistently around 52%. Second, the Lin + Disc model shows a monotonic increase in E_{grid} from 46 to 78 kWh as T_s grows, with the PV fraction dropping from 50.8% to 42.5%. Third, the RMSE chart dramatically illustrates the numerical explosion of the Euler model at $T_s \geq 0.25$ h (reaching orders of 10^{66}), alongside its complete lack of tracking accuracy even prior to mathematical divergence.

In conclusion, $T_s = 0.10$ h applied to the ZOH + Nonlin system represents the optimal trade-off: low computational load, strict adherence to product quality constraints, and maximum renewable energy utilization.

6 PLC Exercise 20 — Conveyor Drive (LAD Program)

Problem Statement

The objective is to operate a three-stage conveyor system (upper, middle, lower) driven by a robot arm, according to sensor data. Parts travel from left to right, detected by sensors X0–X3. The robot supplies one part per cycle when commanded via Y7.

Device	Name	Description
X0	Sensor	Part detected at upper conveyor entry
X1	Sensor	Part detected at upper/middle handoff
X2	Sensor	Part detected at middle/lower handoff
X3	Sensor	Part detected at lower conveyor end (box)
X5	Starting point	Robot is at its home position
X20	PB1	Start push-button on operation panel
Y0/Y1	Upper conveyor	Forward / Reverse
Y2/Y3	Middle conveyor	Forward / Reverse
Y4/Y5	Lower conveyor	Forward / Reverse
Y7	Supply command	Robot supplies one part when ON
Y20	Lamp	Lights when 3 parts counted; resets after 4 s

The seven control specifications are:

1. PB1 (X20) pressed with robot at home (X5) ⇒ Y7 ON; Y7 latched until X5 returns ON.
2. X0 detects part ⇒ `SET Y0` (upper conveyor forward).
3. X1 detects part ⇒ `RST Y0`, `SET Y2` (hand off to middle conveyor).
4. X2 detects part ⇒ `RST Y2`, `SET Y4` (hand off to lower conveyor).
5. X3 detects part ⇒ `RST Y4` (part reaches box).
6. X3 ON and X0 OFF ⇒ M101 set; M101 drives timer T1 (5 s); T1 done ⇒ Y7 ON if X5 ON (automatic re-supply).
7. Counter C0 increments on each X3 pulse; at count 3, Y20 lamp ON and timer T2 (4 s) starts; T2 done ⇒ `RST C0`, `RST Y20`.

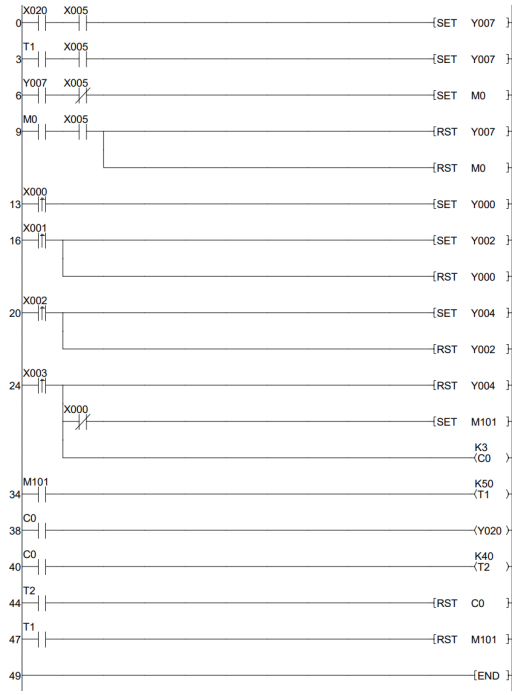

Figure 31: Ladder diagram solution

The program uses SET/RST for all conveyor stages so a stage continues running until the next sensor fires. The supply command Y7 is handled by two parallel paths: one manual (X20 + X5) and one automatic (T1 done + X5), both using the same latch logic to guarantee a complete robot cycle. The part counter (C0, preset = 3) is clocked by the rising edge of X3 and reset by T2 after the 4-second lamp period expires.